SAYED MASHEQUL BARI
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RESEARCH STATEMENT

My research integrates molecular biology wet-lab techniques DNA/RNA extraction, PCR/qPCR, cDNA synthesis, sequencing library preparation, and bacterial culture and identification with bioinformatic analysis spanning genomics, transcriptomics, proteomics, metabolomics and epigenomics. Combining phenotypic and genotypic data, I investigate evolutionary dynamics, immune regulation, host–microbe interactions, and environmental adaptation, working with chromosome-level genome assemblies, short- and long-read bacterial genomes, bulk and single-cell RNA-seq, DNA methylation, shotgun metagenomics, and eDNA datasets to identify genomic signatures, molecular biomarkers, microbial community patterns, and regulatory pathways underlying adaptation. I have authored more than twenty peer-reviewed articles, many as corresponding author, review for several international journals, and mentor graduate and undergraduate students.

EDUCATION

M.S. in Fish Health Management - Sylhet Agricultural University, Bangladesh (2013).
B.Sc. (Hons.) in Fisheries - Sylhet Agricultural University, Bangladesh (2011).

TECHNICAL SKILLS

Molecular lab work

- Experimental design, setup, management of laboratory workflows and biological studies
- DNA extraction, PCR amplification, gel electrophoresis documentation and sequence library preparation
- RNA extraction, cDNA synthesis, RT-PCR, and qPCR quantification and sequence library preparation
- Bacterial culture, biochemical characterization, antibiogram testing, and molecular identification
- Fish parasite morphometric identification, permanent slide preparation, and molecular identification

Bioinformatics skills (See GitHub Repositories, https://github.com/BariSayed)

- **Programming and computational environment**: programming and workflow development in python, r and bash; nextflow pipelines; git/github, conda, pixi and docker; linux hpc and cloud platforms (aws, google cloud)
- **Genomics and variant analysis:** sequence quality assessment and preprocessing; genome alignment and mapping; variant discovery and structural variant analysis
- **Transcriptomics (bulk and single-cell RNA-seq):** read alignment and transcript quantification; differential gene expression analysis; single-cell matrix data clustering, integration, and trajectory analysis.
- **Epigenomic analysis:** genome-wide DNA methylation analysis including quality control, alignment of bisulfite sequencing reads, methylation. Chip-seq data analysis.
- **Metabolomic analysis:** raw data processing and peak detection, metabolite identification and annotation against databases, differential metabolite and pathway enrichment analysis
- **Bacterial whole-genome analysis:** genome assembly (short and long reads) and quality assessment; genome annotation; antimicrobial resistance and gene detection; sequence typing.
- **Metagenomics and microbiome analysis:** amplicon and shotgun metagenomic profiling; taxonomic classification; functional pathway analysis.
- **Mitochondrial genome analysis:** complete mitochondrial genome assembly and characterization, nucleotide composition, codon usage, phylogenetics, and control region characterization.
- **Population and quantitative genomics:** estimation of population differentiation and haplotype diversity; genome-wide association studies; qtl mapping and linkage analysis.
- **Environmental DNA and metabarcoding:** biodiversity assessment, taxonomic assignment, biodiversity and community profiling.
- **Geospatial analysis:** geographic data visualization and spatial mapping using QGIS and ArcGIS.

PUBLICATIONS (denotes corresponding author) * **(See https://barisayed.github.io)**

1. **Bari SM*,** Fuad M, Hossain MJ, Ahammad I, Begum M, Helal MMU. 2025. A meta-analysis of public RNA-Seq data identifies conserved stress responses in Rainbow trout. BMC Genomics. 26: 999. https://doi.org/10.1186/s12864-025-12127-2 **(Q1, IF: 3.9)**
2. Begum M, Barsha KF, Rahman MM, Sarkar MMH, Chowdhury SF, Shormi AS, and **Bari SM***. 2025. Gut microbiome profiling of antibiotic-treated *Mystus cavasius* using culture- based and shotgun metagenomic approaches. Antonie Van Leeuwenhoek. 118(2): 183. https://doi.org/10.1007/s10482-025-02195-5 **(Q3, IF: 2.3)**
3. Hossain MJ, Rasin JH, Altaib H, Shyama IJ, Shawon RAR, Badr Y, Sohidullah M, Sakyi ME, **Bari SM,** Iqbal MM, Rahman MM. 2025. Prevalence and antibiotic resistance patterns of *Vibrio cholerae* isolated from diverse food sources in Khulna, Bangladesh. Food and Humanity. 100653. https://doi.org/10.1016/j.foohum.2025.100653 **(Q3, IF: 3.5)**
4. **Bari SM*,** Ahammad I, Bhattacharjee A, and Sahabuddin AM. 2025. Whole genome sequence data of *Aeromonas diversa* SAU1 isolated from Tilapia (*Oreochromis niloticus*). Data in Brief. 111396. https://doi.org/10.1016/j.dib.2025.111396 **(Q3, IF: 1.4)**
5. **Bari SM***, Marma M, Reja NB, Hossain A, Ahmed SKF, Rayhan, N, Islam MS, Hossain MA, Raman MM, and Alom MS. 2024. In-silico identification of Indian gooseberry (*Emblica officinalis*) compounds inhibiting thermolabile hemolysin from *Vibrio alginolyticus* in Shrimp. Current Computer-Aided Drug Design. 21(2). https://doi.org/10.2174/0115734099342492250120114644 **(Q3, IF: 1.6)**
6. Jahan F, Rahman MH, Ahammad B, Khandaker M, Rahman MZ, Rasel M, Nafis R, Shaila A, **Bari SM**, Chandra MB. 2025. *Argulus japonicus*: A concerning health issue for *Channa punctata* collected from south-western Bangladesh. Veterinary Medicine and Science. 11(2). 11:e70250. https://doi.org/10.1002/vms3.70250 **(Q1, IF: 1.7)**
7. **Bari SM***, Reza NB, Marma M, Ahmed SKF, Hossain MA, Jabed MN, Hossain M.A., Monzoor M., and Alom M.S. 2025. Computational analysis of *Allium sativum* compounds to identify thermolabile hemolysin inhibitors against *Vibrio alginolyticus* in Shrimp. Journal of Advanced Biotechnology and Experimental Therapeutics. 8(1): 01-20. https://doi.org/10.5455/jabet.2025.01 **(Q4, CS: 2.2)**
8. **Bari SM*,** Islam MM, Amina A, Khatun MM, and Sahabuddin AM. 2024. Molecular identification, histopathology, and antibiotic susceptibility profiling of *Aeromonas veronii* isolated from *Oreochromis niloticus* in Bangladesh. Veterinary Medicine and Science, 10(6): e70103. https://doi.org/10.1002/vms3.70103 **(Q1, IF: 1.7)**
9. **Bari SM*,** Rahman MM, Amina A, Nadia ZM, Shawon RAR, Rahman M.M., Masum M.M. and Habib K.A. 2024. Morphological and molecular identification of *Euclinostomum heterostomum* (Trematoda: Clinostomidae) from spotted snakehead *Channa punctata* in Bangladesh. Systematic Parasitology. 101:62. https://doi.org/10.1007/s11230-024-10175-y **(Q3, IF: 1.2)**
10. **Bari SM*,** Hossen MI, Islam MR, and Helal MMU. 2024. Genome-wide identification and characterization of Interleukin-18 gene family in Rainbow trout (*Oncorhynchus mykiss*). Journal of Advanced Biotechnology and Experimental Therapeutics. 7(2): 275-289. https://doi.org/10.5455/jabet.2024.d23 **(Q4, CS: 2.2)**
11. Hossen, MK, Bari SM, Barman PP, Roy R, and Das PK. 2022. Application of Python OpenCV to detect contour of shapes and colour of a real image. International Journal of Novel Research in Computer Science and Software Engineering, 9, 20-25. https://doi.org/10.5281/zenodo.6576264
12. Deb A, Chandan S, Roy P, Hossain MI, **Bari SM**. 2021. Inland aquaculture and fish health management: a case study of Sylhet district in Bangladesh. Aquaculture Studies. 21: 129-137. http://doi.org/10.4194/2618-6381-v21_3_05 **(Q4, CS: 2.2)**

IN REVIEW

1. **Bari SM*,** Khanum H, Reza NB, Marma M, Amina A. Habib KA. 2026. A checklist of helminth parasites in freshwater fishes of Bangladesh. Veterinary Medicine and Sciences **(Accepted) (Q1, IF: 1.7)**
2. **Bari SM***Marma M, Khan SN, Paul DC**,** Habib KA, Karim MM. 2026. First complete mitochondrial genome of Asian seabass (*Lates calcarifer*) from Bangladesh reveals Indian Ocean lineage affiliation and evolutionary relationships within Latidae. Aquaculture Report **(In Review) (Q1, IF: 3.8)** http://dx.doi.org/10.2139/ssrn.7133980
3. **Bari SM***, Imon MRI, Javeed MN, Mitu S, Islam T, Hossain MA, Islam MS 2026. Immunoinformatics design of a novel chimeric multiepitope vaccine against bacterial diseases in Pangas catfish (*Pangasius pangasius*). PLOS ONE. (**In Review**) **(Q1, IF: 3.2)**
4. Sarker BK, Begum M, **Bari SM**, Sultana N, Islam S, Chowdhury F, Kabir MA, Rabbane MG. 2026. Gut microbiota composition and nutritional quality of flathead grey mullet (*Mugil cephalus*) from riverine and gher aquaculture systems in coastal Bangladesh. Environmental Microbiology (**In Review**) (**Q1, IF: 4.0**)
5. **Bari SM***, Rahman Z, Mostafa F, Hossain M, Rahman MA, Begum M, and Helal MMU. 2026. Single-cell meta-analysis reveals cell-type-specific transcriptional responses to cardiac injury in zebrafish. BMC Genomics. (**In Review**) (**Q1, IF: 3.9**).
6. **Bari1 SM*,** Rahman MA, Zahangir MM, Begum M, Habib KA, Helal MMU. 2026. QTL-rich cluster analysis identifies genomic regions for disease resistance and growth in rainbow trout (*Oncorhynchus mykiss*). Aquaculture Report (**In Review**) (**Q1, IF: 4.0**).

7. **Bari SM*,** Mostafa F, Hossain S, Mamun MAA, Karim MM. 2026. Pangenome analysis of human and aquaculture Streptococcus agalactiae identifies integrative conjugative elements as major carriers of antimicrobial resistance asymmetry. BMC Microbiology (**In Review**) (**IF: 5.4**)
8. Helal MMU*, Israfil M, **Bari SM***. 2026. Shared and stress-specific DNA methylation programs across biotic and abiotic stresses in rice (*Oryza sativa*): a multi-study meta-analysis. BMC Plant Biology (**In Review**) (**Q1, IF: 4.0**).

CONFERENCE PRESENTATION AND AWARDS

- **Co-chair, Fish Disease Session,** Fisheries Society of Bangladesh Young Fisheries Students Congress (FSB YFSC-2025) held on 8th January 2026, Dhaka University, Dhaka, Bangladesh.
- **Best Oral Presenter**, Fish Disease session, 10th Biennial Conference of BFRF, December 2025, organized by Bangladesh Fisheries Research Forum (BFRF), Dhaka, Bangladesh
- **Best Oral Presenter**, Computational Biology and Bioinformatics session, International Conference of Biotechnology on Health and Agriculture (ICBHA), September 26-27, 2025, organized by Global Network of Bangladeshi Biotechnologists (GNOBB), Dhaka, Bangladesh
- **Best Oral Presenter**, Marine Biotechnology Session, International Conference on Biotechnology, June 20–21, 2025, organized by BRAC University, Dhaka, Bangladesh
- **Oral Presenter**, Fish Disease Session, Scientific Conference on Sustainable Aquaculture and Fisheries, April 29–30, 2025, organized by Chattogram Veterinary and Animal Sciences University, Chattogram, Bangladesh
- **Oral Presenter**, Aquaculture and Disease Session, International Fisheries and Aquaculture Conference, June 8, 2024, Department of Aquaculture, Rajshahi University, Rajshahi, Bangladesh
- **Oral Presenter**, International Conference on Advanced Agricultural Research, May 24–25, 2024, organized by Sylhet Agricultural University, Sylhet, Bangladesh.

PROFESSIONAL MEMBERSHIP

- Member, American Society of Microbiology (ASM)
- Member, European Association of Fish Pathologists (EAFP)
- Member, Global Network of Bangladesh Biotechnologists (GNOBB), Bangladesh
- Member, International Society of Computational Biology (ISCB)
- Member, American Fisheries Society (AFS), United States
- Member, Bangladesh Fisheries Research Forum (BFRF), Bangladesh
- Member, Zoological Society of Bangladesh (ZSB), Bangladesh
- Member, Bangladesh Society of Microbiologists (BSM), Bangladesh
- Member, Krishibid Institute Bangladesh (KIB), Bangladesh
- Member, Sylhet Agricultural University Alumni Association (SAUAA), Bangladesh

LEADERSHIP EXPERIENCE

- Vice-president, Sylhet Agricultural University Blood Donation Club, Sylhet Agricultural University, Sylhet, Bangladesh, Jan 2010 - Dec 2010
- Vice-president, Fisheries Student Association, Sylhet Agricultural University, Sylhet, Bangladesh, Jan 2011 - Dec 2011
- Vice-president, Sylhet Agricultural University Science Club, Sylhet Agricultural University, Sylhet, Bangladesh, Jan 2012 - Dec 2012

RESEARCH SUPERVISION

- Humaira Tasnuva Boichi (Master's supervision, 2026), Diet Metabarcoding and Gut Microbiome Analysis in Invasive Sucker Fish Across Buringanga River and Aquaculture Systems.
- Ratri Rani Roy (Master's supervision, 2026), *Streptococcus agalactiae* isolated from commercial tilapia farms in Bangladesh: unravelling antibiotic resistance genes, virulence determinants, and integrative and conjugative elements (ices) through oxford nanopore technology (ONT) sequencing a one health perspective
- Md. Muttakinul Islam (Master's supervision, 2025), Whole genome sequence of *Aeromonas* spp. collected from disease infected Parshe Fish (Grey Mullet).
- Meamaching Marma, (Master's supervision, 2025), Characterization of the complete mitochondrial genome of Asian seabass (*Lates calcarifer*) from Bangladesh.
- Nafees Bin Reza (Master's supervision, 2025), Temporal hematological, inflammatory responses and immune gene expression of Nile tilapia (*Oreochromis niloticus*) following experimental *Aeromonas veronii* infection.
- Md. Abdul Jabbar (Undergraduate Thesis, 2024), Morphometric and molecular identification of helminth parasites collected from infected snakeheads (*Channa punctata*) in Bangladesh.